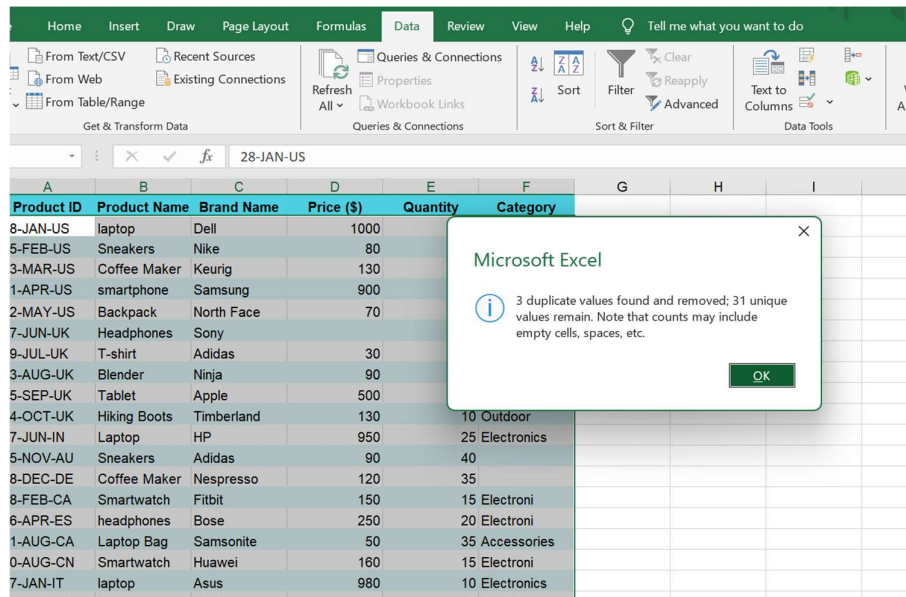


Excel Assignment 2 – Data Exploration

1.Removing Duplicates:

The duplicate data in the dataset is removed using “remove duplicate” option in the data tool.



2.Handling Missing Values:

The given data set had some missing values in “Price” and “Category” column. By selecting Blanks in the filters, the missing values can be filtered.

2.1 In price column

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
07-JUN-UK	Headphones	Sony		45	Electronics
14-AUG-US	Camping Tent	Coleman		10	Outdoor
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion

To find the missing price details, the formula is used,

“=IFERROR(IF(ISBLANK(D2),AVERAGEIFS(\$D\$2:\$D\$32,\$F\$2:\$F\$32,F2,\$B\$2:\$B\$32,B2),D2),AVERAGEIF(\$F\$2:\$F\$32,F2,\$D\$2:\$D\$32))”

Average Price based on Category and Product Name is calculated using above formula for imputing the values, however in case of non availability of sufficient data (wherever error was thrown), average Price based on Category alone is imputed. In the given data set,

- The price of Headphones and Sunglasses are imputed based on average of Category and Product Name.
- The price of Camping Tent is imputed based on average of Category alone.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price
JAN-US	Laptop	Dell	1000	30	Electronics	1000
FEB-US	Sneakers	Nike	80	15	Fashion	80
MAR-US	Coffee Maker	Keurig	130	40	Kitchen	130
APR-US	Smartphone	Samsung	900	25	Electronics	900
MAY-US	Backpack	North Face	70	20		70
JUN-UK	Headphones	Sony		45	Electronics	250
JUL-UK	T-Shirt	Adidas	30	5	Fashion	30
AUG-UK	Blender	Ninja	90	35	Kitchen	90
SEP-UK	Tablet	Apple	500	50	Electronics	500
OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130
JUN-IN	Laptop	HP	950	25	Electronics	950
NOV-AU	Sneakers	Adidas	90	40		90

2.2 In Category column:

In category column we have 4 missing values.

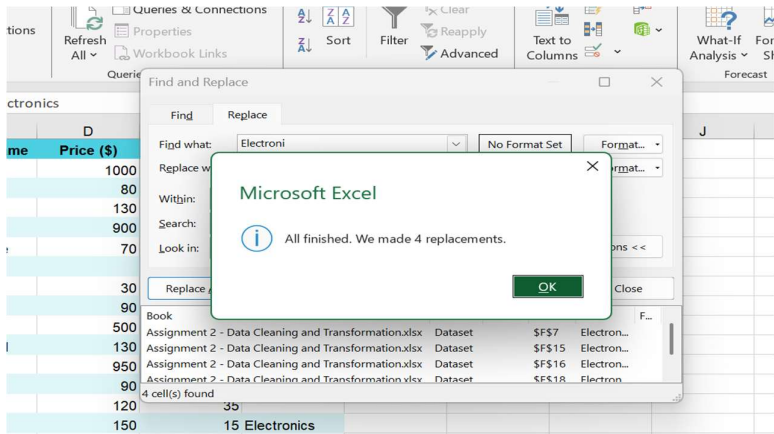
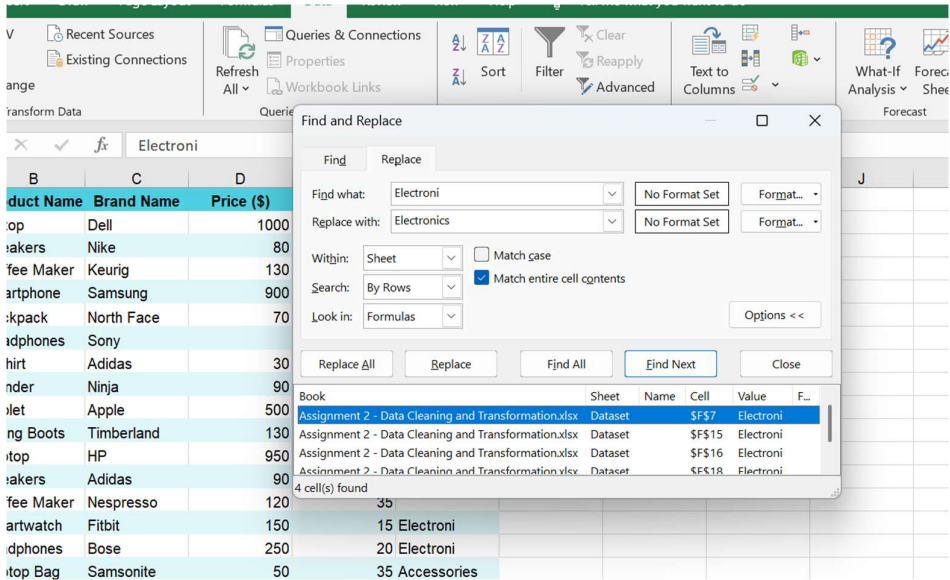
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
22-MAY-US	Backpack	North Face	70	20	
25-NOV-AU	Sneakers	Adidas	90	40	
08-DEC-DE	Coffee Maker	Nespresso	120	35	
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	

Using the “IF” function the missing category values are imputed as “Unknown”.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Imputed Category
18-JAN-US	Laptop	Dell	1000	30	Electronics	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion	Fashion
13-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics	Electronics
12-MAY-US	Backpack	North Face	70	20		Unknown
17-JUN-UK	Headphones	Sony		45	Electronics	Electronics
19-JUL-UK	T-Shirt	Adidas	30	5	Fashion	Fashion
13-AUG-UK	Blender	Ninja	90	35	Kitchen	Kitchen
15-SEP-UK	Tablet	Apple	500	50	Electronics	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics	Electronics
15-NOV-AU	Sneakers	Adidas	90	40		Unknown
18-DEC-DE	Coffee Maker	Nespresso	120	35		Unknown
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics	Electronics

3. Correcting Inconsistent Data:

Using the “**PROPER**” function the product name are arranged in a proper text format. Some misspellings were available in the category column. Using “**FIND AND REPLACE**” the spellings are corrected.



4. Splitting and Merging Data:

Using “Text to column” in data tool the "Product ID" column is split into separate columns like "Manufacturing Date" and "Country Code".

Convert Text to Columns Wizard - Step 1 of 3

The Text Wizard has determined that your data is Delimited.
If this is correct, choose Next or choose the data type that best describes your data.

Original data type
Choose the file type that best describes your data:
☒ Delimited - Characters such as commas or tabs separate each field.
☐ Fixed width - Fields are aligned in columns with spaces between each field.

Preview of selected data:

Product ID	Manufacturing Date
28-JAN-US	
15-FEB-US	
03-MAR-US	
11-APR-US	
22-MAY-US	
07-JUN-UK	
19-JUL-UK	
23-AUG-UK	
05-SEP-UK	
14-OCT-UK	
17-JUN-IN	
25-NOV-AU	
08-DEC-DE	
18-FEB-CA	
16-APR-ES	
21-AUG-CA	
20-AUG-CN	
27-JAN-IT	

Convert Text to Columns Wizard - Step 2 of 3

This screen lets you set the delimiters that your data contains. You can see how your text is affected in the preview below.

Delimiters
☐ Tab
☐ Semicolon
☐ Comma
☐ Space
☒ Other: [none]

Test qualifier: [none]

Data preview

Product ID	Manufacturing Date
28	JAN US
15	FEB US
03	MAR US
11	APR US
22	MAY US
07	JUN UK
19	JUL UK
23	AUG UK
05	SEP UK
14	OCT UK
17	JUN IN
25	NOV AU
08	DEC DE
18	FEB CA
16	APR ES
21	AUG CA
20	AUG CN
27	JAN IT

Convert Text to Columns Wizard - Step 3 of 3

This screen lets you select each column and set the Data Format.

Column data format
☐ General
☒ Text
☐ Date: DMY
☐ Do not import column (skip)

Destination: \$A\$2

Data preview

Product ID	Manufacturing Date
28	JAN US
15	FEB US
03	MAR US
11	APR US
22	MAY US
07	JUN UK
19	JUL UK
23	AUG UK
05	SEP UK
14	OCT UK
17	JUN IN
25	NOV AU
08	DEC DE
18	FEB CA
16	APR ES
21	AUG CA
20	AUG CN
27	JAN IT

Excel Spreadsheet

	A	B	C	D
1	Product ID	Manufacturing Date	Month	Country code
2	28-JAN-US	28 JAN	US	
3	15-FEB-US	28 FEB	US	
4	03-MAR-US	28 MAR	US	
5	11-APR-US	28 APR	US	
6	22-MAY-US	28 MAY	US	
7	07-JUN-UK	28 JUN	UK	
8	19-JUL-UK	28 JUL	UK	
9	23-AUG-UK	28 AUG	UK	
10	05-SEP-UK	28 SEP	UK	
11	14-OCT-UK	28 OCT	UK	
12	17-JUN-IN	28 JUN	IN	
13	25-NOV-AU	28 NOV	AU	
14	08-DEC-DE	28 DEC	DE	
15	18-FEB-CA	28 FEB	CA	
16	16-APR-ES	28 APR	ES	
17	21-AUG-CA	28 AUG	CA	
18	20-AUG-CN	28 AUG	CN	
19	27-JAN-IT	28 JAN	IT	
20	01-MAR-IJK	28 MAR	IJK	

The "Brand Name" and "Product Name" columns are merged into one column named "Product Brand" using the formula “=CONCATENATE(C2," ",B2)”.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Product brand
JAN-US	Laptop	Dell	1000	30	Electronics	Dell Laptop
FEB-US	Sneakers	Nike	80	15	Fashion	Nike Sneakers
MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Keurig Coffee Maker
APR-US	Smartphone	Samsung	900	25	Electronics	Samsung Smartphone
MAY-US	Backpack	North Face	70	20		North Face Backpack
JUN-UK	Headphones	Sony		45	Electronics	Sony Headphones
JUL-UK	T-Shirt	Adidas	30	5	Fashion	Adidas T-Shirt
AUG-UK	Blender	Ninja	90	35	Kitchen	Ninja Blender
SEP-UK	Tablet	Apple	500	50	Electronics	Apple Tablet
OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Timberland Hiking Boots
NOV-IN	Laptop	HP	950	25	Electronics	HP Laptop
DEC-AU	Sneakers	Adidas	90	40		Adidas Sneakers
JAN-DE	Coffee Maker	Nespresso	120	35		Nespresso Coffee Maker
FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Fitbit Smartwatch
MAR-ES	Headphones	Bose	250	20	Electronics	Bose Headphones
APR-CA	Laptop Bag	Samsonite	50	35	Accessories	Samsonite Laptop Bag
MAY-CN	Smartwatch	Huawei	160	15	Electronics	Huawei Smartwatch

5. Number Formatting:

- The "Manufacturing Date" column is changed in to date format using the formula

“=DATEVALUE(CONCATENATE(I2," ",J2))”.

- The Price column is converted to dollar format (\$) using the Currency option under number formatting option in Home Tab.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price range	Category	Manufacturing Date	Manufacturing Date	Country
JAN-US	Laptop	Dell	1000	30	Electronics	\$1,000.00	Electronics	28 JAN		US
FEB-US	Sneakers	Nike	80	15	Fashion	\$80.00	Fashion	15 FEB	15-02-2026	US
MAR-US	Coffee Maker	Keurig	130	40	Kitchen	\$130.00	Kitchen	3 MAR	03-03-2026	US
APR-US	Smartphone	Samsung	900	25	Electronics	\$900.00	Electronics	11 APR	11-04-2026	US
MAY-US	Backpack	North Face	70	20		\$70.00	unknown	22 MAY	22-05-2026	US
JUN-UK	Headphones	Sony		45	Electronics	\$297.86	Electronics	7 JUN	07-06-2026	UK
JUL-UK	T-Shirt	Adidas	30	5	Fashion	\$30.00	Fashion	19 JUL	19-07-2026	UK
AUG-UK	Blender	Ninja	90	35	Kitchen	\$90.00	Kitchen	23 AUG	23-08-2026	UK
SEP-UK	Tablet	Apple	500	50	Electronics	\$500.00	Electronics	5 SEP	05-09-2026	UK
OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	\$130.00	Outdoor	14 OCT	14-10-2026	UK
NOV-IN	Laptop	HP	950	25	Electronics	\$950.00	Electronics	17 JUN	17-06-2026	IN
DEC-AU	Sneakers	Adidas	90	40		\$90.00	unknown	25 NOV	25-11-2026	AU
JAN-DE	Coffee Maker	Nespresso	120	35		\$120.00	unknown	8 DEC	08-12-2026	DE
FEB-CA	Smartwatch	Fitbit	150	15	Electronics	\$150.00	Electronics	18 FEB	18-02-2026	CA

6. Conditional Formatting:

Using the **Condition formatting** options in the “Home” page data bars are applied for price column. The cells containing “Electronics” in the category column is highlighted using conditional formatting option “**Highlight cells rule**”.

29-09-2026										
A	B	C	D	E	F	G	H	I	J	K
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Imputed Price (\$)	Imputed Category	Manufacturing Date	Country code	Product
JAN-US	Laptop	Dell	1000	30	Electronics	\$1,000.00	Electronics	28-01-2026	US	Dell Laptop
FEB-US	Sneakers	Nike	80	15	Fashion	\$80.00	Fashion	15-02-2026	US	Nike Sneakers
MAR-US	Coffee Maker	Keurig	130	40	Kitchen	\$130.00	Kitchen	03-03-2026	US	Keurig Coffee Maker
APR-US	Smartphone	Samsung	900	25	Electronics	\$900.00	Electronics	11-04-2026	US	Samsung Smartphone
MAY-US	Backpack	North Face	70	20		\$70.00	Unknown	22-05-2026	US	North Face Backpack
JUN-UK	Headphones	Sony		45	Electronics	\$250.00	Electronics	07-06-2026	UK	Sony Headphones
JUL-UK	T-Shirt	Adidas	30	5	Fashion	\$30.00	Fashion	19-07-2026	UK	Adidas T-Shirt
AUG-UK	Blender	Ninja	90	35	Kitchen	\$90.00	Kitchen	23-08-2026	UK	Ninja Blender
SEP-UK	Tablet	Apple	500	50	Electronics	\$500.00	Electronics	05-09-2026	UK	Apple Tablet
OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	\$130.00	Outdoor	14-10-2026	UK	Timberland Hiking Boots
JUN-IN	Laptop	HP	950	25	Electronics	\$950.00	Electronics	17-06-2026	IN	HP Laptop
NOV-AU	Sneakers	Adidas	90	40		\$90.00	Unknown	25-11-2026	AU	Adidas Sneakers
DEC-DE	Coffee Maker	Nespresso	120	35		\$120.00	Unknown	08-12-2026	DE	Nespresso Coffee Maker
FEB-CA	Smartwatch	Fitbit	150	15	Electronics	\$150.00	Electronics	18-02-2026	CA	Fitbit Smartwatch
APR-ES	Headphones	Bose	250	20	Electronics	\$250.00	Electronics	16-04-2026	ES	Bose Headphones
AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	\$50.00	Accessories	21-08-2026	CA	Samsonite Laptop Bag
AUG-CN	Smartwatch	Huawei	160	15	Electronics	\$160.00	Electronics	20-08-2026	CN	Huawei Smartwatch
JAN-IT	Laptop	Asus	980	10	Electronics	\$980.00	Electronics	27-01-2026	IT	Asus Laptop
MAR-UK	Sunglasses	Oakley	150	15	Fashion	\$150.00	Fashion	01-03-2026	UK	Oakley Sunglasses
AUG-US	Camping Tent	Coleman		10	Outdoor	#DIV/0!	Outdoor	14-08-2026	US	Coleman Camping Tent
MAY-RU	Camera	Nikon	700	50	Electronics	\$700.00	Electronics	14-05-2026	RU	Nikon Camera
JAN-CA	Microwave	Panasonic	80	20	Kitchen	\$80.00	Kitchen	09-01-2026	CA	Panasonic Microwave
JUL-BR	Fitness Tracker	Xiaomi	150	30		\$150.00	Unknown	19-07-2026	BR	Xiaomi Fitness Tracker

16-04-2026										
A	B	C	D	E	F	G	H	I	J	K
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Imputed Price (\$)	Imputed Category	Manufacturing Date	Country code	Product Brand
28-JAN-US	Laptop	Dell	1000	30	Electronics	\$1,000.00	Electronics	28-01-2026	US	Dell Laptop
15-FEB-US	Sneakers	Nike	80	15	Fashion	\$80.00	Fashion	15-02-2026	US	Nike Sneakers
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	\$130.00	Kitchen	03-03-2026	US	Keurig Coffee Maker
11-APR-US	Smartphone	Samsung	900	25	Electronics	\$900.00	Electronics	11-04-2026	US	Samsung Smartphone
22-MAY-US	Backpack	North Face	70	20		\$70.00	Unknown	22-05-2026	US	North Face Backpack
07-JUN-UK	Headphones	Sony		45	Electronics	\$250.00	Electronics	07-06-2026	UK	Sony Headphones
19-JUL-UK	T-Shirt	Adidas	30	5	Fashion	\$30.00	Fashion	19-07-2026	UK	Adidas T-Shirt
23-AUG-UK	Blender	Ninja	90	35	Kitchen	\$90.00	Kitchen	23-08-2026	UK	Ninja Blender
05-SEP-UK	Tablet	Apple	500	50	Electronics	\$500.00	Electronics	05-09-2026	UK	Apple Tablet
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	\$130.00	Outdoor	14-10-2026	UK	Timberland Hiking Boots
17-JUN-IN	Laptop	HP	950	25	Electronics	\$950.00	Electronics	17-06-2026	IN	HP Laptop
25-NOV-AU	Sneakers	Adidas	90	40		\$90.00	Unknown	25-11-2026	AU	Adidas Sneakers
08-DEC-DE	Coffee Maker	Nespresso	120	35		\$120.00	Unknown	08-12-2026	DE	Nespresso Coffee Maker
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	\$150.00	Electronics	18-02-2026	CA	Fitbit Smartwatch
16-APR-ES	Headphones	Bose	250	20	Electronics	\$250.00	Electronics	16-04-2026	ES	Bose Headphones
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	\$50.00	Accessories	21-08-2026	CA	Samsonite Laptop Bag
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	\$160.00	Electronics	20-08-2026	CN	Huawei Smartwatch
27-JAN-IT	Laptop	Asus	980	10	Electronics	\$980.00	Electronics	27-01-2026	IT	Asus Laptop
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	\$150.00	Fashion	01-03-2026	UK	Oakley Sunglasses
14-AUG-US	Camping Tent	Coleman		10	Outdoor	#DIV/0!	Outdoor	14-08-2026	US	Coleman Camping Tent
14-MAY-RU	Camera	Nikon	700	50	Electronics	\$700.00	Electronics	14-05-2026	RU	Nikon Camera
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	\$80.00	Kitchen	09-01-2026	CA	Panasonic Microwave
19-JUL-BR	Fitness Tracker	Xiaomi	150	30		\$150.00	Unknown	19-07-2026	BR	Xiaomi Fitness Tracker